Kevin Kuister

Zusammenfassung: Lemke - Howson

Lemke - Howson

$$L(x) = \{ i \mid x_i = 0 \} \cup \{ j \mid (Q^T x)_j \geq 1 \}$$

$$L(y) = \{ i \mid y_i = 0 \} \cup \{ j \mid (P y)_j \geq 1 \}$$

$$y \geq 0, \quad x \geq 0$$

	A	B
A	(3,1)	(2,2)
B	(2,3)	(3,1)

$$P = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} \Rightarrow P y \leq 1 \quad \begin{cases} 3y_3 + 2y_4 \leq 1 \Rightarrow y_4 = \frac{1}{2} - \frac{3}{2} y_3 \; (=) \; (0 \mid \frac{1}{2}), (\frac{1}{3} \mid 0) \\ 2y_3 + 3y_4 \leq 1 \Rightarrow y_4 = \frac{1}{3} - \frac{2}{3} y_3 \; (=) \; (0 \mid \frac{1}{3}), (\frac{1}{2} \mid 0) \end{cases}$$

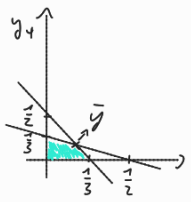
$$Q^T = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} \Rightarrow Q^T x \leq 1 \quad \begin{cases} x_1 + 3x_2 \leq 1 \Rightarrow x_2 = \frac{1}{3} - \frac{1}{3} x_1 \; (=) \; (0 \mid \frac{1}{3}), (1 \mid 0) \\ 2x_1 + x_2 \leq 1 \Rightarrow x_2 = 1 - 2x \; (=) \; (0 \mid 1), (\frac{1}{2} \mid 0) \end{cases}$$

$$3y_3 + 2y_4 + r_1 = 1$$

$$2y_3 + 3y_4 + r_2 = 1$$

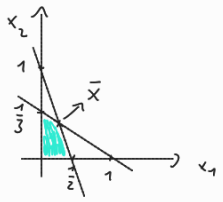
$$x_1 + 3x_2 + s_3 = 1$$

$$2x_1 + x_2 + s_4 = 1$$



$\bar{y} = (\frac{1}{5}, \frac{1}{5})$

$(0,0); (\frac{1}{3}, 0); (0, \frac{1}{3}), (\frac{1}{5}, \frac{1}{5})$



$\bar{x} = (\frac{1}{5}, \frac{1}{5})$

$(0,0); (\frac{1}{2}, 0); (0, \frac{1}{3}); (\frac{1}{5}, \frac{1}{5})$

X	L(x)
$(0, \frac{1}{3})$	1,3

$x_1 = 0$

$x_1 + 3x_2 + s_3 = 0$

$(=) \; 0 + 3 \cdot \frac{1}{3} + s_3 = 1$

$\Downarrow$

$= 0$

Punkte-Kombination x-y:

	x	y	L(x)	L(y)	L(x) ∪ L(y)
A	(0,0)	(0,0)	1,2	3,4	1,2,3,4
B	$(0, \frac{1}{3})$	(0,0)	1,3	3,4	1,3,4
C	$(\frac{2}{5}, \frac{1}{5})$	(0,0)	3,4	3,4	3,4
D	$(\frac{1}{2}, 0)$	(0,0)	2,4	3,4	2,3,4
F		$(0, \frac{1}{2})$	1,2	2,3	1,2,3
G		⋮	1,3	2,3	1,2,3
H		⋮	3,4	2,3	2,3,4
I	⋮	⋮	2,4	2,3	2,3,4
J		$(\frac{1}{5}, \frac{1}{5})$	1,2	1,2	1,2
K		⋮	1,3	1,2	1,2,3
L		⋮	3,4	1,2	1,2,3,4
M		⋮	2,4	1,2	1,2,4
N		$(\frac{1}{3}, 0)$	1,2	1,4	1,2,4
O		⋮	1,3	1,4	1,3,4
P		⋮	3,4	1,4	1,3,4
Q		⋮	2,4	1,4	1,2,4

$V_1 \Rightarrow$ 1. Label

	x_1	x_2	s_3	s_4	b
s_3	1	3	1	0	1
s_4	2	1	0	1	1

Start: V_1

Pivot: Kleinster Wert für $\frac{s_i}{b_i}$ $\Rightarrow$ $\frac{1}{2}$

	x_1	x_2	s_3	s_4	b
s_3	1	3	1	0	1
s_4	2	1	0	1	1

Gauß: Basisvektor $\Rightarrow \begin{matrix} 0 \\ 1 \end{matrix}$

	x_1	x_2	s_3	s_4	b
s_3	1	3	1	0	1
s_4	2	1	0	1	1
s_3	0	5/2	1	-1/2	1/2
x_1	1	1/2	0	1/2	1/2

Tausch: $1 \xrightarrow{x} 4$ $\Rightarrow$ Weiter mit y_4

	y_3	y_4	r_1	r_2	b
r_1	3	2	1	0	1
r_2	2	3	0	1	1
r_1	5/3	0	1	-2/3	1/3
y_4	2/3	1	0	1/3	1/3

$1 \xrightarrow{x} 4$

$4 \xrightarrow{y} 2$ $\Rightarrow$ Weiter mit x_2

$\rightarrow$ bis Zyklus $\Rightarrow$ fertig

$V_1 \Rightarrow$ 1. Label

	x_1	x_2	s_3	s_4	b
s_3	1	3	1	0	1
s_4	2	1	0	1	1
s_3	0	5/2	1	-1/2	1/2
x_1	1	1/2	0	1/2	1/2
x_2	0	1	2/5	-1/5	1/5
x_1	1	0	-1/5	3/5	2/5

$1 \xrightarrow{x} 4$

$4 \xrightarrow{y} 2$

$2 \xrightarrow{x} 3$

$3 \xrightarrow{y} 1$

	y_3	y_4	r_1	r_2	b
r_1	3	2	1	0	1
r_2	2	3	0	1	1
r_1	5/3	0	1	-2/3	1/3
y_4	2/3	1	0	1/3	1/3
y_3	1	0	3/5	-2/5	1/5
y_4	0	1	-1/5	3/5	1/5

Lösung:

$x = (\frac{2}{5}, \frac{1}{5})$

$y = (\frac{1}{5}, \frac{1}{5})$

$\frac{x_i}{x_1 + x_2}$ NORMAL

$p = (\frac{2}{3}, \frac{1}{3})$

$q = (\frac{1}{2}, \frac{1}{2})$